Technical Workflow Specification: Teaching Materials Generator (n8n)

This document describes the complete technical design of the current version of the workflow: its architecture, nodes, data flow, expressions, credentials, validation, deployment and testing. It replaces the Ollama-based setup described in earlier notes, because the current workflow uses Google Gemini for the AI step and adds a validation branch and PDF output.

1. System overview

Item	Detail
Purpose	Turn one lesson topic and learning objective into a complete, differentiated teaching package
Platform	n8n, self-hosted on a local PC (http://localhost:5678)
Trigger	n8n Form Trigger (web form)
AI	6 AI Agent nodes, each with its own Google Gemini Chat Model sub-node
Validation	If node checks that all 12 sections (A to L) are present
Output	PDF attached to an email sent to the teacher
Failure path	Alert email to the workflow owner when a section is missing
Public access	ngrok tunnel to the local n8n, with the form published at its Production URL
Total nodes	20 (1 trigger, 6 agents, 6 model sub-nodes, 7 processing and output nodes)

The package contains: A Learning Objectives and Success Criteria, B Teacher Explanation, C Student Handout, D Differentiated Learning Versions (Beginner, Intermediate, Advanced / Challenge), E Dyslexia-Friendly Learning Version, F Examples and Activities, G

Practice Exercises, H Discussion Questions, I Quiz and Answer Key, J Summary, K Visual-Content Prompts, L Accessibility and Teacher Guidance.

2. Architecture and execution flow

```
[On form submission]
        |
        v
[Core Content] -> [Differentiated Versions] -> [Dyslexia Version]
        -> [Activities and Exercises] -> [Quiz and Summary] -> [Visuals and
Guidance]
        |
        v
[Edit Fields]  (builds one text field: package)
        |
        v
[If]  (are all 12 section headings present?)
   |-- true  --> [Markdown] -> [Wrap Page] -> [Convert HTML to PDF] -> [Send
a message]
   |-- false --> [Send a message1]  (alert email to the owner)
```

Each of the six agents has a Google Gemini Chat Model connected to its Chat Model input. The agents run one after another, not in parallel, so that each model call stays within the rate limits and each node has the previous output available.

2.1 Node inventory

#	Node name	Type	Role
1	On form submission	n8n Form Trigger	Collects the teacher's inputs
2	Core Content	AI Agent	Sections A, B, C (zero-shot)
3	Differentiated Versions	AI Agent	Section D (one-shot)
4	Dyslexia Version	AI Agent	Section E (few-shot)
5	Activities and Exercises	AI Agent	Sections F, G, H (zero-shot)
6	Quiz and Summary	AI Agent	Sections I, J (zero-shot)
7	Visuals and Guidance	AI Agent	Sections K, L (zero-shot)

#	Node name	Type	Role
8 to 13	Google Gemini Chat Model to Chat Model5	Chat Model sub-nodes	One model per agent
14	Edit Fields	Set	Combines the six outputs into `package`
15	If	Condition	Validates that all 12 sections exist
16	Markdown	Markdown	Converts the package text to HTML
17	Wrap Page	Set	Wraps the HTML in a styled page, field `page`
18	Convert HTML to PDF	HTTP Request	Calls a PDF conversion API
19	Send a message	Gmail	Sends the PDF to the teacher
20	Send a message1	Gmail	Sends the failure alert to the owner

3. Data contract

The data changes shape as it moves through the workflow. Each node reads earlier results by node name, so the names must never be changed.

Stage	Field produced	Content
Form	Subject, Grade Level, Lesson Topic, Learning Objective, Lesson Duration, Curriculum or Standards, Language, Available Resources, Specific Learner Needs, Assessment Requirements, Email	Teacher inputs (text)
Each agent	`output`	Markdown text for its sections
Edit Fields	`package`	All six outputs joined in order, with a title and an info line

Stage	Field produced	Content
If	none (routing only)	Passes the same item to the true or false branch
Markdown	`html`	HTML version of `package`
Wrap Page	`page`	Complete HTML page with styling
Convert HTML to PDF	binary field (see 5.9)	The PDF file
Gmail	none	Sent email with the PDF attached

4. Credentials

Credential	Used by	How it is set up
Google Gemini (PaLM) API	The 6 Gemini Chat Model nodes	Create an API key at Google AI Studio, then in n8n choose Google Gemini(PaLM) Api, keep the default host https://generativelanguage.googleapis.com, and paste the key
Gmail OAuth2	Both Gmail nodes	Google Cloud project with the Gmail API enabled, an OAuth client (Web application) with the redirect URI http://localhost:5678/rest/oauth2-credential/callback, and your address listed under Test users
PDF service key	Convert HTML to PDF	The API key or token of the PDF conversion service, entered in the HTTP Request authentication settings, as that service's documentation describes

Credentials are stored inside n8n and are not part of an exported workflow file.

5. Node configuration

5.1 On form submission (n8n Form Trigger)

- **Authentication:** None
- **Form Title:** Teaching Materials Generator
- **Form Description:** Enter your lesson once and receive the full teaching package by email.
- **Option, Respond When:** Form Is Submitted (the form closes at once while the workflow keeps running)

Field Name	Element Type	Required
Subject	Text	Yes
Grade Level	Text	Yes
Lesson Topic	Text	Yes
Learning Objective	Text	Yes
Lesson Duration	Text	Yes
Curriculum or Standards	Text	No
Language	Text	Yes
Available Resources	Text	No
Specific Learner Needs	Text	No
Assessment Requirements	Text	No
Email	Email	Yes

5.2 The six AI Agent nodes

All six agents share the same structure.

- **Source for Prompt (User Message):** Define below
- **Prompt (User Message):** Expression mode. The Teacher Inputs Block comes first, then the agent's task text. Every field is read with an expression such as `{{ $('On form submission').item.json['Lesson Topic'] }}`.

- **Options > System Message:** the Shared System Message, plus the sentence "Keep every section concise: use short bullet points and brief paragraphs, and avoid repeating information."
- **Settings tab:** Retry On Fail ON, Max Tries 3, Wait Between Tries 10000 ms (gives Gemini time to recover from rate-limit errors)

The exact prompts and the shared system message are in the document Prompt Library: AI Teaching Materials Generator.

Agent	Prompting technique	Sections	Why
Core Content	Zero-shot	A, B, C	Clear task, no example needed
Differentiated Versions	One-shot (Magnets example)	D	One example sets structure and tone for three levels
Dyslexia Version	Few-shot (Magnets and Plants examples)	E	Several examples show real dyslexia-friendly adaptation
Activities and Exercises	Zero-shot	F, G, H	Clear list of required items
Quiz and Summary	Zero-shot	I, J	Clear list of required items
Visuals and Guidance	Zero-shot	K, L	Clear list of required items

5.3 Google Gemini Chat Model (one per agent)

- **Credential:** the Google Gemini credential
- **Model:** a Flash model chosen from the dropdown (fast, with a free tier), the same in all six nodes
- **Options > Sampling Temperature:** 0.3 (more accurate and consistent science content)

5.4 Edit Fields

- **Mode:** Manual Mapping
- **Name:** `package` (typed text) | **Type:** String | **Value** (Expression):

```
# Teaching Package: {{ $('On form submission').item.json['Lesson Topic'] }}

Subject: {{ $('On form submission').item.json.Subject }} | Grade: {{ $('On
form submission').item.json['Grade Level'] }} | Duration: {{ $('On form
submission').item.json['Lesson Duration'] }} | Language: {{ $('On form
submission').item.json.Language }}

{{ $('Core Content').item.json.output }}

{{ $('Differentiated Versions').item.json.output }}

{{ $('Dyslexia Version').item.json.output }}

{{ $('Activities and Exercises').item.json.output }}

{{ $('Quiz and Summary').item.json.output }}

{{ $('Visuals and Guidance').item.json.output }}
```

This node exists because each agent outputs only its own part, and the following nodes need one single field. The validation check is not placed in this node, because the field `package` does not exist yet while this node is running.

5.5 If (validation)

- **Condition, Value 1** (Expression):

```
{{ ['## A.','## B.','## C.','## D.','## E.','## F.','## G.','## H.','##
I.','## J.','## K.','## L.'].every(h => $json.package.includes(h)) }}
```

- **Operator:** Boolean, is true
- **true output:** goes to Markdown
- **false output:** goes to Send a message1

The check confirms that all twelve section headings exist. It cannot judge whether the facts are correct, so a human review is still required (section 8).

5.6 Markdown

- **Mode:** Markdown to HTML
- **Markdown:** `{{ $json.package }}`

- **Destination Key:** `html`
- **Options > Tables:** ON

5.7 Wrap Page

- **Mode:** Manual Mapping
- **Name:** `page` | **Type:** String | **Value** (Expression):

```
<html><head><meta charset="utf-8"><style>body{font-family:Arial,sans-serif;font-size:14px;line-height:1.6;margin:30px} h1,h2,h3{margin-top:24px} table{border-collapse:collapse} td,th{border:1px solid #999;padding:6px} </style></head><body>{{ $json.html }}</body></html>
```

- **Include Other Input Fields:** OFF

The sans-serif font and spacing follow the readability guidance for the dyslexia-friendly content.

5.8 Convert HTML to PDF (HTTP Request)

This node sends the HTML page to an online PDF conversion service and receives a PDF file back.

- **Method:** POST
- **URL:** the conversion endpoint of the PDF service (it begins with `https://api.pdf...` in the workflow)
- **Authentication:** the key or token required by that service
- **Body:** the HTML from `{{ $json.page }}`, sent in the parameter name that the service expects (for example `html`)
- **Options > Response > Response Format:** File
- **Put Output in Field:** `data`

The exact URL, authentication method and parameter names depend on the provider, so confirm them against its documentation. If the node is changed to another provider, only this node needs to change.

5.9 Send a message (Gmail, success path)

- **Credential:** Gmail OAuth2
- **Resource / Operation:** Message / Send

- **To** (Expression): `{{ $('On form submission').item.json.Email }}`
- **Subject** (Expression): `Teaching package: {{ $('On form submission').item.json['Lesson Topic'] }}`
- **Email Type:** Text
- **Message:** a short text that says the package is attached and asks the teacher to check the facts, quiz answers and answer key before use, because the content is AI-generated
- **Options > Attachments > Attachment Field Name:** `data` (the binary field that holds the PDF)
- **Options > Append n8n attribution:** OFF

5.10 Send a message1 (Gmail, failure path)

- **To:** the owner's own email address
- **Subject:** Teaching package incomplete
- **Message:** A section is missing. Please run the form again.

6. Deployment and public access

The form is opened by teachers through a public link created with ngrok.

1. **Start ngrok** in its own Command Prompt window: `ngrok http 5678`. Keep it open and note the Forwarding address (https://...ngrok-free.dev).
2. **Start n8n** in a second window with the public address:

```
set WEBHOOK_URL=https://YOUR-ADDRESS.ngrok-free.dev/
set N8N_PROXY_HOPS=1
n8n start
```

3. **Publish** the workflow in n8n.
4. Open the **On form submission** node, choose the **Production URL** tab, and copy the link (it looks like https://YOUR-ADDRESS.ngrok-free.dev/form/...).
5. Send this link to teachers. The first time they open it, ngrok may show a warning page where they click Visit Site.

While the form is in use, the PC, the ngrok window and the n8n window must stay running. If ngrok gives a different address next time, repeat steps 2 to 5 with the new address, or use a fixed free domain from the ngrok dashboard.

7. Error handling and reliability

Risk	Control
Gemini rate limit or temporary error	Retry On Fail on every agent (3 tries, 10 s wait)
One section missing from the output	If node routes to the alert email, and the teacher package is not sent
Wrong node name in an expression	Keep node names exactly as in section 2.1
Form waits too long	Respond When set to Form Is Submitted
Gmail or PDF service failure	Check the Executions tab for the red node and read its error message
Whole workflow fails	Optional: create a separate workflow with an Error Trigger and a Gmail node, then select it under workflow Settings > Error Workflow

8. Quality, privacy and security

- **Human review is required.** The AI can make factual mistakes. The teacher must check the science facts, quiz answers and answer key before using the materials. The email message says so.
- **No invented standards.** If no curriculum standard is entered, the system message tells the model to state assumptions and never to invent standards.
- **Data sent to Google.** Lesson inputs are sent to the Gemini API. On the free tier, Google states that content may be used to improve its products. Do not enter student names or personal data in the form.
- **API keys.** Keep the Gemini key and the PDF key private. Do not post them or show them in screenshots. If a key is exposed, delete it and create a new one.
- **Public form.** Anyone with the link can submit the form and start a run, which uses model quota and sends an email from the owner's Gmail account. Share the link only with trusted people, and close ngrok or unpublish the workflow to stop access.
- **n8n sign-in.** The same public address also reaches the n8n sign-in page, so use a strong n8n password.

9. Test plan

Test	Input	Expected result
Normal run	Science, Grade 3, States of matter, objective, 45 minutes, English, valid email	All nodes green, PDF arrives by email
Different level	Same topic with Grade 1 and Grade 5	Language, support and task difficulty differ clearly between the two packages
Optional fields blank	Leave standards, resources, learner needs and assessment empty	Package is created, and assumptions are stated in the content
Missing section	Remove one agent's output (for testing only)	If node goes to the false branch and the alert email arrives
Invalid email	Wrong address in the form	Form blocks it, or the Gmail node shows an error
Public link	Open the Production URL in a private window	Form opens (after the ngrok warning page) and the submission starts a run

For the learner-level comparison deliverable, run the form twice with different Grade Level values and compare the Beginner, Intermediate, Advanced and Dyslexia-Friendly sections side by side.

10. Maintenance and export

- To export, open the three-dot menu in the workflow and choose Download, which saves a JSON file. To import, create a new workflow and choose Import from file.
- Credentials are not included in the file. The new user must create their own Gemini, Gmail and PDF service credentials.
- After any change, save the workflow and publish it again so the live form uses the new version.
- Models and free-tier limits change over time, so check the model dropdown and the limits in Google AI Studio from time to time.

11. Known limitations

- A run takes several minutes because six model calls run one after another.
- The section check only confirms that headings exist. It does not verify content quality.
- Free-tier rate limits may slow or block runs when many teachers submit the form at once.
- The public link works only while the PC, n8n and ngrok are running.
- The PDF step depends on an external online service, so the workflow needs internet access and a valid key for that service.